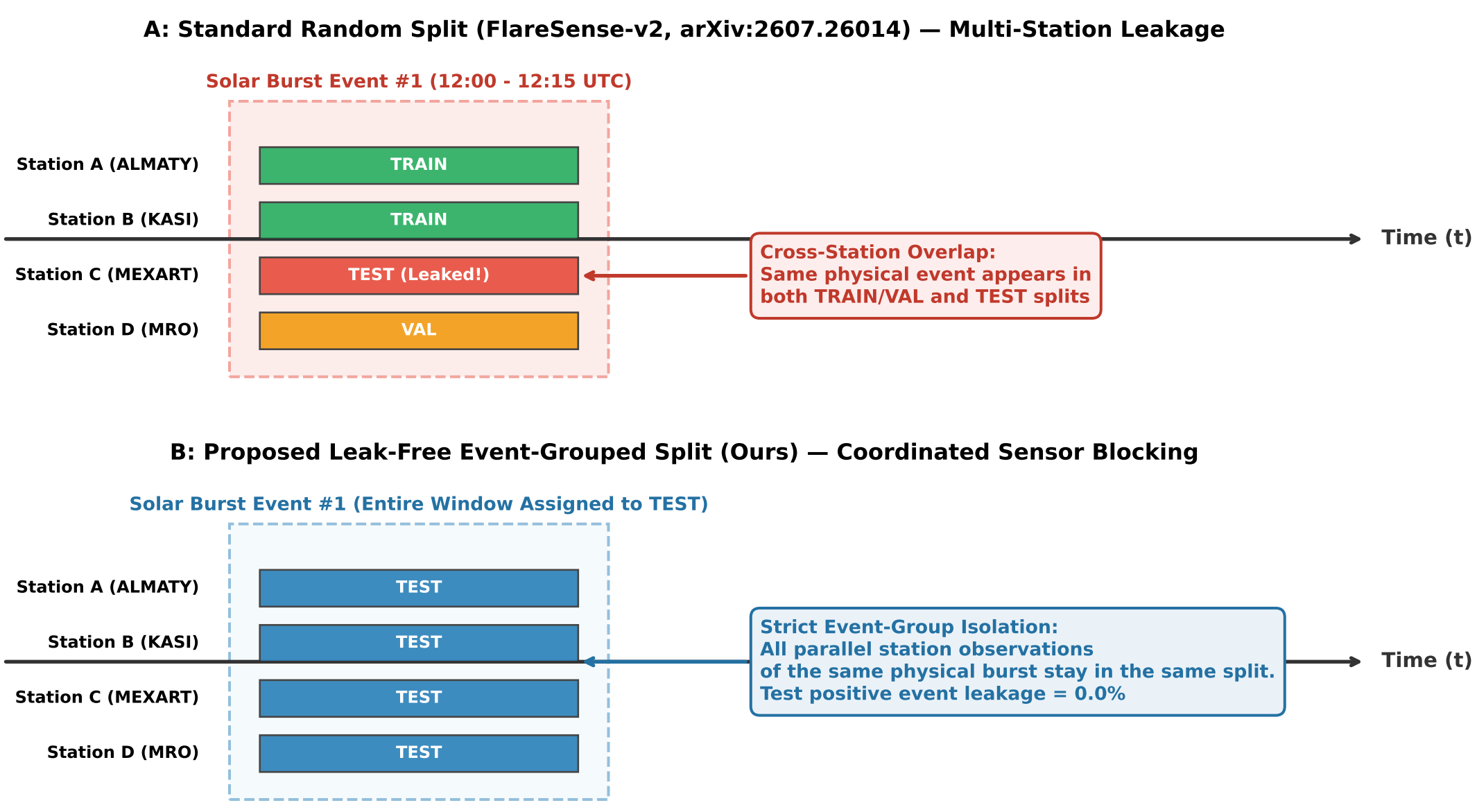
A: Standard Random Split (FlareSense-v2, arXiv:2607.26014) — Multi-Station Leakage
Solar Burst Event #1 (12:00 - 12:15 UTC)
Station A (ALMATY)
TRAIN
Station B (KASI)
TRAIN
Station C (MEXART)
TEST (Leaked!)
Station D (MRO)
VAL
Time (t)
Cross-Station Overlap:
Same physical event appears in
both TRAIN/VAL and TEST splits
B: Proposed Leak-Free Event-Grouped Split (Ours) — Coordinated Sensor Blocking
Solar Burst Event #1 (Entire Window Assigned to TEST)
Station A (ALMATY)
TEST
Station B (KASI)
TEST
Station C (MEXART)
TEST
Station D (MRO)
TEST
Time (t)
Strict Event-Group Isolation:
All parallel station observations
of the same physical burst stay in the same split.
Test positive event leakage = 0.0%